Catastrophic Forgetting in Sequential Thermal Anti-UAV Detection

Scale-Conditioned Gradient Imbalance as a Candidate Forgetting Mechanism

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Abstract

Counter-UAV systems based on thermal infrared detection must stay accurate as operational datasets evolve, yet sequential fine-tuning causes catastrophic forgetting of prior tasks, a problem not previously quantified in this domain. This continual-learning study measures the stability-plasticity trade-off in YOLOMG, a YOLOv5-based detector run as a single thermal-infrared stream (motion channel disabled), trained sequentially across three anti-UAV benchmarks of rising scale difficulty: Anti-UAV-RGBT, Anti-UAV410, and CST Anti-UAV. Naive fine-tuning on CST yields a Forgetting Measure of −0.605 against the Stage 1 ceiling (90% capability loss), −0.572 of it in Stage 3 alone, whereas knowledge distillation from a frozen teacher limits forgetting to FM = -0.033 ± 0.004 across three seeds (95% retention). Per-stratum analysis shows large-target detection collapsing to near-zero within the first epoch, despite an inter-stage cosine similarity of 0.987 over the gradient-updated weights, pointing to a scale-conditioned gradient imbalance, rather than weight drift, as a candidate mechanism. Scale-Stratified Herding (SSH), a 300-exemplar buffer balanced across four UAV size strata, roughly halves the forgetting (FM = $-0.605 \rightarrow -0.311$) and keeps large-target detection non-zero; an ablation attributes the gain to the scale stratification rather than the herding, with random-stratified replay performing at least as well.

Keywords

Continual Learning, Scale-Distribution Shift, Catastrophic Forgetting, Thermal Anti-UAV Detection, Knowledge Distillation, Gradient Starvation

Github Repository

https://github.com/GiangNguyen06/UvA-Masters-Thesis-UAV

1 Introduction

The proliferation of commercial drones has created urgent demand for counter-unmanned-aerial-vehicle (C-UAV) surveillance systems capable of reliable detection across diverse operational environments [46]. Thermal infrared (TIR) imaging is increasingly favoured over visible-spectrum cameras for this role: it operates independently of ambient lighting, remains effective against low-visibility backgrounds, and provides a distinct radiometric signature for airborne targets [13, 49]. Deep learning detectors, particularly YOLO-family architectures [15], perform strongly on individual TIR anti-UAV benchmarks [9]. But practical C-UAV systems must stay accurate as new datasets emerge, sensors are upgraded, and the threat profile shifts toward smaller, harder-to-detect targets.

The core technical challenge this creates is catastrophic forgetting [8, 24]: when a neural network is sequentially fine-tuned on a new dataset, the weight updates that improve performance on the new task interfere with representations learned for previous ones, causing abrupt performance degradation on earlier tasks [17, 25]. The field of continual learning (CL) exists to manage this stability-plasticity trade-off [40]. Rehearsal methods store a subset of past training data and interleave it during subsequent training [2, 5, 32]; regularisation methods penalise changes to important parameters [17, 22]; gradient projection methods constrain new-task gradient updates to subspaces orthogonal to previous tasks' gradients [7, 34]; and knowledge distillation (KD) methods use a frozen teacher model to constrain the updated model's output distribution [12, 20]. Shmelkov et al. [36] introduced distillation-based CL for general object detection, showing it prevents forgetting when new classes are added. None of these approaches have been studied in the thermal anti-UAV domain, where distributional shifts involve not only sensor modality but dramatic changes in target scale.

Existing thermal anti-UAV benchmarks capture complementary facets of this problem in isolation. Anti-UAV-RGBT [14] provides 318 paired RGB-TIR video streams of standard-scale drones; Anti-UAV410 [13] contains 410 purely thermal sequences spanning a wide range of drone sizes; and CST Anti-UAV [43] focuses on extreme tiny targets in cluttered thermal scenes, with 87.9% tiny and 11.9% small bounding boxes and effectively no large targets. No prior work has chained these datasets into a single model's training history and measured the resulting stability-plasticity trade-off. Prior continual learning work has also not examined gradient starvation [30], a phenomenon in which dominant features suppress gradient signals for weaker but relevant ones, as a driver of forgetting after distributional shift. This is the research gap this work addresses.

We investigate the following hypothesis and three research questions using YOLOMG [9], a dual-modal YOLOv5-based detector (run here with its motion channel disabled), trained sequentially on Anti-UAV-RGBT (Stage 1), Anti-UAV410 (Stage 2), and CST Anti-UAV (Stage 3). All three datasets are thermal infrared, so the input modality is held constant across the curriculum; the shifts under study are the moderate cross-dataset domain shift of Stage 2 and the extreme scale-distribution shift of Stage 3.

H1: The magnitude of catastrophic forgetting of Task 1 under sequential fine-tuning is governed by the type of distribution shift, specifically, the scale-distribution shift of Stage 3 produces forgetting at least an order of magnitude larger than the cross-dataset thermal domain shift absorbed by knowledge distillation in Stage 2.

RQ1: Does knowledge distillation preserve Task 1 performance during Task 2 training under domain shift? Specifically, does the KD loss term in Stage 2 (Anti-UAV410) training prevent the model from forgetting its Anti-UAV-RGBT detection capability, despite the thermal cross-dataset domain shift between the two benchmarks?

RQ2: How does scale shift characterise the pattern of catastrophic forgetting during Task 3 training? Specifically, does the extreme size distribution of CST Anti-UAV (87.9% tiny, 11.9% small, 0% large) produce stratum-specific forgetting in which large-target detection collapses disproportionately fast, and how is this reflected in per-stratum mean Average Precision (mAP), the Forgetting Measure, parameter drift, and the gradient signal reaching each detection head?

RQ3: Does re-injecting scale-stratified exemplars via replay mitigate the Stage 3 forgetting? A 300-exemplar buffer that selects exemplars across four UAV size strata (tiny, small, normal, large) by iCaRL-style greedy herding restores the gradient signal for the under-represented strata during Stage 3, and is compared against naive fine-tuning and a random-stratified control that isolates the effect of the stratification.

In brief, sequential training appears to induce substantial and measurable forgetting. Knowledge distillation acts as an effective safeguard during Stage 2. The Stage 3 results point to a scale-induced forgetting pattern whose evidence is, tentatively, more consistent with a scale-conditioned gradient imbalance than with outright weight overwriting. Scale-Stratified Herding (SSH) roughly halves this forgetting, which suggests that scale-stratified replay can offset the imbalance.

The remainder of this work is organised as follows. Section 2 surveys related work on continual learning, thermal UAV detection, and knowledge distillation. Section 3 describes the experimental design, datasets, and methods. Section 4 presents the empirical results. Section 5 interprets the findings and discusses limitations. Section 6 concludes with implications and future directions.

2 Related Work

No prior work has combined a continual learning protocol with thermal anti-UAV detection across multiple real operational datasets, nor characterised scale-conditioned gradient imbalance, a form of gradient starvation, as a candidate driver of forgetting under scale-distribution shift. This section situates the research within four bodies of literature that together delimit this gap.

2.1 Thermal Anti-UAV Detection

Counter-UAV surveillance using thermal infrared imaging has received growing attention due to the sensor's independence from ambient lighting and its distinct radiometric signature for airborne targets. Zhao et al. [46] provide a broad survey of vision-based anti-UAV detection and tracking, identifying the shift from visible to thermal modalities as a key trend driven by operational requirements at night and in degraded visibility. Jiang et al. [14] introduced Anti-UAV-RGBT, a paired RGB-TIR benchmark of 318 video sequences that remains the most widely used dual-modality anti-UAV evaluation resource. Huang et al. [13] extended this with Anti-UAV410, a purely thermal benchmark of 410 sequences spanning a wider range of acquisition conditions and drone distances. Both datasets focus on standard-scale commercial quadcopters; the more recent CST Anti-UAV [43] shifts attention to extreme tiny targets in cluttered thermal scenes, creating the near-complete scale inversion relative to Anti-UAV-RGBT that motivates the forgetting analysis in this research. Detection in such conditions shares challenges with the broader infrared small target detection (IRSTD) literature, where methods such as DNA-Net [18] use dense nested attention to preserve sub-16-pixel target features that deep pooling layers would otherwise discard.

On the detector side, YOLO-family architectures [15] have become the dominant choice for real-time anti-UAV detection because of their favourable speed-accuracy balance. Guo et al. [9] extended YOLOv5 with a pixel-level motion difference channel (YOLOMG), achieving drone-to-drone detection by fusing appearance and motion at three feature pyramid scales (P3, P4, P5) [21]. Zhu et al. [49] proposed integrating evidential uncertainty into the detection-to-tracking pipeline, providing a reference point for detection confidence calibration under occlusion. Domain adaptation from RGB to thermal and across weather conditions has been explored by Zhang et al. [45] and Zheng et al. [47], but these works address single-step adaptation rather than sequential multi-dataset training. None of these lines of work measures catastrophic forgetting of a previous dataset after adapting to a new one.

2.2 Continual Learning

The instability of neural networks under sequential training has been studied since McCloskey and Cohen [24] documented catastrophic interference in connectionist models. Goodfellow et al. [8] provided the first systematic empirical investigation of this phenomenon in modern gradient-based networks, examining how activation functions, optimisers, and task relationships modulate the degree of forgetting. Mermillod et al. [25] formalised the stability-plasticity dilemma: a model must retain prior knowledge while acquiring new capabilities, and optimising for one degrades the other.

Van de Ven et al. [39] formalise three CL scenarios: task-incremental learning (task identity known at test time), domain- incremental learning (same class set, changing input distribution), and class-incremental learning (new classes arrive without task identity). Our three-stage curriculum is domain-incremental: the target class (UAV) is fixed across all stages and the input modality is held constant at thermal infrared (only the IR stream is used), while the scale regime shifts from standard-scale to wide-range to extreme tiny-target.

Continual learning is one of several responses to distribution shift. Domain generalisation and one-shot domain adaptation aim to transfer a model to a new distribution without necessarily preserving source-task performance, while parameter-efficient adaptation—low-rank adapters (LoRA) and related PEFT methods—has recently been applied to sequential and domain-incremental settings, including object detection [10, 19, 42]. This work adopts continual learning because the operational goal is *cumulative* retention across a fixed sequence of real anti-UAV datasets, for which catastrophic forgetting is the defining failure mode [40]. Replay is preferred over a low-rank weight constraint because, as Section 4.4 shows, the failure is a missing *scale-conditioned gradient*: replay re-injects that signal directly, whereas constraining weight movement does not supply it. Parameter-efficient continual adaptation nonetheless remains a promising alternative and is revisited as future work.

The continual learning literature organises mitigation methods into four families [3, 27, 48]; Wang et al. [40] find that rehearsal consistently outperforms regularisation-only approaches as task dissimilarity increases. *Regularisation* methods such as Elastic Weight Consolidation (EWC) [17] penalise changes to parameters important for previous tasks; this limits weight change but not the decay of an unreinforced feature, whose gradient can vanish without any weight change. *Architecture* methods (Progressive Neural Networks [33], PackNet [23]) allocate or mask per-task parameters to avoid forgetting by construction, but scale poorly to long task sequences. *Gradient-projection* methods, Orthogonal Gradient Descent [7] and Gradient Projection Memory [34], project new-task updates orthogonal to previous-task directions, eliminating gradient *interference*. That assumption (forgetting as overwriting) only partly holds under the scale-conditioned gradient imbalance examined in Section 4.4, where large-target capability degrades because the new distribution supplies no positive gradient for it, not because it is overwritten. *Rehearsal* methods interleave stored exemplars with new data: iCaRL [32] selects exemplars by greedy herding to the class mean (the strategy adapted for Scale-Stratified Herding here), Chaudhry et al. [2] show even tiny memories help, GEM [22] constrains updates not to increase loss on stored exemplars, and Dark Experience Replay [1] adds logit distillation. Deng et al. [5] show rehearsal bounds forgetting when the replayed distribution covers the original in feature space, motivating the scale-stratified design of the SSH buffer. Distillation methods use a frozen copy of the previous model as a teacher [12]; Learning without Forgetting (LwF) [20] penalises divergence from the teacher's output distribution on new-task inputs without storing any previous data.

2.3 Incremental Object Detection

Applying continual learning to object detectors is non-trivial because the detection loss couples classification, localisation, and objectness objectives. Shmelkov et al. [36] established the foundational approach: a distillation loss on the old-class output logits of a Faster R-CNN prevents catastrophic forgetting when new classes are added, while the detection loss on new-class ground truth drives plasticity. Wang et al. [41] refined this with Elastic Response Distillation (ERD), which selectively weights the distillation signal by the quality of each detection response, achieving state-of-the-art incremental detection on MS COCO. Joseph et al. [16] approached incremental object detection with a meta-learning strategy that reshapes model gradients to minimise forgetting while maximising knowledge transfer across incrementally added classes. Both Shmelkov et al., ERD, and Joseph et al. study class-incremental detection: new classes arrive sequentially on the same image distribution. Song et al. [38] study domain-incremental detection, the setting where the class set is fixed but the image distribution changes across tasks, which is closer to the protocol used in this research. Their non-exemplar method learns a compact domain bias module per task, but does not quantify per-stratum forgetting or examine gradient starvation as a mechanism. No prior work in incremental object detection has (a) applied a CL protocol to a chain of real thermal anti-UAV datasets or (b) identified scale-distribution shift between tasks as a primary driver of forgetting.

2.4 Gradient Starvation and Scale Distribution Shift

Pezeshki et al. [30] characterised gradient starvation as a learning proclivity in which a dominant feature receives the majority of the gradient signal, suppressing updates to weaker but relevant features. This phenomenon has been studied in the context of spurious correlation and shortcut learning in single-task settings, but has not previously been linked to catastrophic forgetting in object detection or anti-UAV surveillance. The connection this research draws is that when a new task's training distribution lacks a feature class that was dominant in previous tasks (specifically, large-target bounding boxes in Anti-UAV-RGBT), the positive (target-matched) gradient signal for that feature class largely vanishes during new-task training. Singh and Davis [37] demonstrated in the single-task setting that CNNs are not robust to scale changes between training and test time: a model trained on large objects underperforms systematically on small ones and vice versa. In the continual learning setting studied here, this scale sensitivity manifests across tasks rather than within a single evaluation: large-target features learned on Anti-UAV-RGBT receive no reinforcing positive gradient signal during CST Anti-UAV fine-tuning and decay as a result. Because the weights encoding large-target representations are not actively overwritten, the resulting forgetting is dissociated from weight-change magnitude, as captured by the inter-stage cosine similarity analysis in Section 4.4. This distinguishes gradient starvation from the gradient interference studied by methods such as OGD [7] and GPM [34]: those methods project conflicting gradients, but cannot project gradients that do not exist. This scale-conditioned imbalance is therefore a failure mode that existing regularisation, gradient projection, and distillation methods are not designed to address, since none restores a positive gradient signal that the new distribution does not provide.

3 Methodology

This research follows a sequential empirical evaluation design. A single detector is trained across three anti-UAV datasets in order, and the change in detection performance on earlier data is measured after each subsequent training stage. The design is motivated by the continual learning literature [40], which formalises this as a stability-plasticity problem: the model must retain prior knowledge (stability) while acquiring new capabilities (plasticity). The protocol corresponds to the domain-incremental learning scenario defined by Van de Ven et al. [39]: the target class (UAV) is fixed throughout all three stages, while the input distribution changes with each new dataset. Unlike prior anti-UAV studies that evaluate each dataset in isolation, this research chains Anti-UAV-RGBT, Anti-UAV410, and CST Anti-UAV into a shared training history so that forgetting can be measured directly. Figure 1 summarises the three-stage curriculum.

3.1 Backbone: YOLOMG

The detector is YOLOMG [9], a YOLOv5-based architecture [15] extended with a second input channel. The first channel is the infrared appearance frame. The second channel is a pixel-level motion difference map (called mask32) computed within a five-frame window, using the three frames $t-2$, t, $t+2$ (Appendix E), after global motion compensation (GMC). GMC removes camera ego-motion using the Enhanced Correlation Coefficient algorithm [6], leaving only object-relative motion as the residual signal. The two channels are fused by concatenation in the early convolutional layers. YOLOMG outputs predictions at three detection heads (P3, P4, P5) for small, medium, and large objects. Its anchors are small and dataset-specific (re-clustered for tiny UAVs; largest 26×14 px), not the YOLOv5 defaults. The full model has 318 layers and 335 named parameter tensors.

YOLOMG was selected because the initial scope included motion-based detection, with the mask32 channel intended to complement the infrared stream [9, 46, 49]. As the scope narrowed to a continual

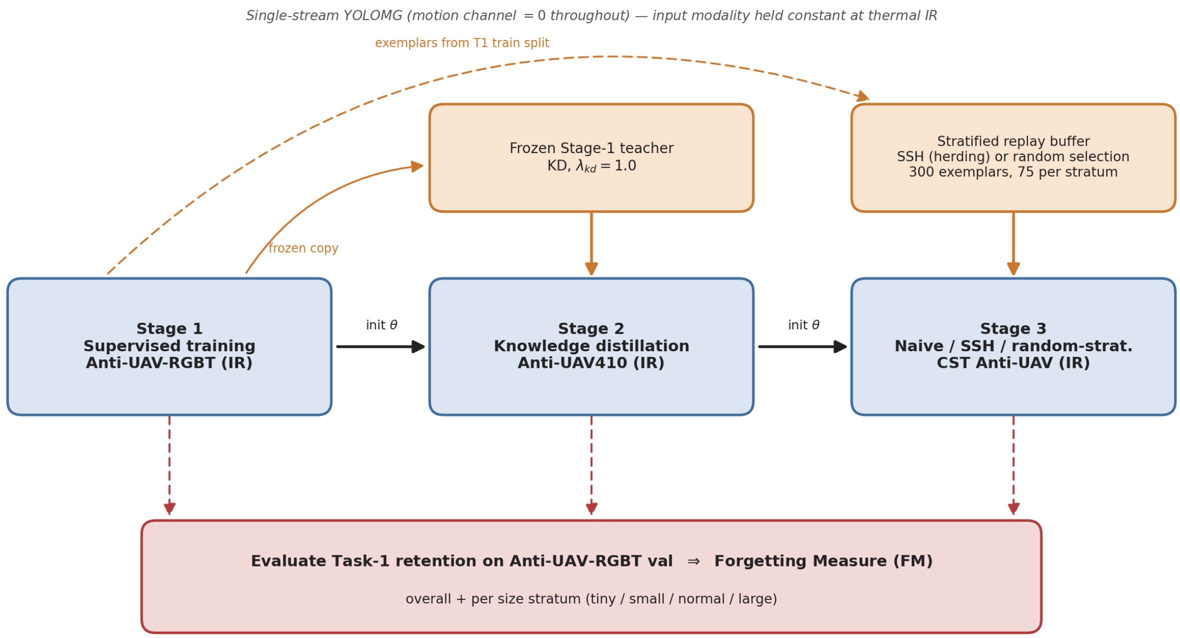

Figure 1: Overview of the three-stage continual-learning curriculum. A single YOLOMG detector, run as a single thermal-infrared stream with the motion channel zeroed, is trained sequentially: Stage 1 supervised on Anti-UAV-RGBT, Stage 2 with knowledge distillation from the frozen Stage 1 teacher on Anti-UAV410, and Stage 3 (naive fine-tuning, Scale-Stratified Herding replay, or the random-stratified control) on CST Anti-UAV. The replay buffer is drawn from the Anti-UAV-RGBT training split (300 exemplars, 75 per size stratum). After each stage, Task-1 retention is evaluated on the Anti-UAV-RGBT validation split to compute the Forgetting Measure, overall and per size stratum.

learning protocol, the motion channel was ruled out: Anti-UAV-RGBT [14] and Anti-UAV410 [13] are appearance-based tracking datasets whose inter-frame UAV displacement is too small for the GMC-based mask32 to yield an informative signal. Rather than switch architecture mid-project, adding a confound to the analysis, the motion channel was held at a zero tensor throughout, making the model functionally a single-stream YOLOv5 appearance detector while keeping the architecture consistent across the curriculum.

3.2 Stage 1: Supervised Training on Anti-UAV-RGBT

Stage 1 trains YOLOMG from ImageNet-pretrained weights on the infrared stream of Anti-UAV-RGBT [14] using fully supervised detection. Ground-truth bounding box annotations are available for the IR stream directly, so no domain adaptation is needed at this stage. The model is trained for 100 epochs using SGD with momentum 0.937 and weight decay 5×10^{-4}, with a cosine learning rate schedule decaying from an initial rate of 10^{-2}. The best checkpoint by validation mAP@0.5 is retained and used to initialise Stage 2. Stage 1 performance sets the T1 ceiling against which all subsequent Forgetting Measure values are computed.

3.3 Stage 2: Knowledge Distillation on Anti-UAV410

Stage 2 fine-tunes the Stage 1 checkpoint on Anti-UAV410 [13], a larger thermal infrared dataset that shares the same large-to-normal scale regime as Stage 1 but covers a wider range of acquisition conditions. The risk at this stage is that fine-tuning on Anti-UAV410 destroys the representations built for Anti-UAV-RGBT. To prevent this, Stage 2 uses knowledge distillation [12, 20]: the frozen Stage 1 checkpoint acts as a teacher, and the fine-tuning student is penalised for diverging from the teacher's output distribution.

The total training loss is:

$$L_{\text{total}} = L_{\text{det}} + \lambda_{\text{kd}} \cdot L_{\text{kd}} \tag{1}$$

where L_{det} is the standard YOLOMG detection loss (box regression, objectness, and classification) and L_{kd} is the mean squared error between the student and teacher prediction grids at the three detection heads. The weight is fixed at $\lambda_{\text{kd}} = 1.0$, weighting detection and distillation equally. This follows convention: both Li and Hoiem [20] and Shmelkov et al. [36] use unit weighting as their default. Equal weighting is a neutral baseline, biasing the optimisation toward neither plasticity (lower λ) nor stability (higher λ) without task-specific tuning data. Stage 2 is run three times with different random seeds (42, 123, 999) to obtain confidence intervals on the Forgetting Measure and the peak T2 mAP.

3.4 Stage 3: Naive Fine-tuning on CST Anti-UAV

Stage 3 fine-tunes the best Stage 2 checkpoint on CST Anti-UAV [43] without any memory protection mechanism. This is the naive baseline against which future mitigation strategies will be compared. The purpose of Stage 3 is diagnostic. It quantifies how much forgetting the scale shift from Anti-UAV410 (49% normal/large) to CST (99.8% tiny/small) causes, and examines the parameter-space changes that accompany it. Training runs for the number of epochs needed to reach peak T3 mAP, after which T1 is evaluated.

3.5 Replay Design: Scale-Stratified Herding

The gradient-starvation interpretation developed in the Stage 3 analysis motivates the design of Scale-Stratified Herding (SSH). Standard iCaRL herding [32] selects exemplars whose penultimate-layer embeddings are closest to the class mean, but treats all samples uniformly. In the anti-UAV setting, the T1 training split is imbalanced across size strata (Table 6). A flat herding buffer would replicate this imbalance rather than guarantee coverage of every stratum, so the strata that are scarce in T1 would receive little gradient protection during Stage 3.

SSH partitions the buffer capacity equally across four size strata (tiny, small, normal, large) defined by bounding box area in pixels. Exemplars are drawn from the Anti-UAV-RGBT training split (148,368 UAV-present frames), kept disjoint from the validation set used to compute the FM. Within each stratum, greedy herding selects the 75 samples closest to the stratum mean embedding, for a total buffer of 300 exemplars. This small fixed memory follows the low-budget rehearsal convention of iCaRL [32] and the finding that even tiny episodic memories curb forgetting [2]; the equal 75-per-stratum split is the neutral choice that removes the source-distribution imbalance. As the qualitative conclusion (stratified replay reduces forgetting relative to naive fine-tuning) does not depend on the exact budget, a systematic buffer-size sweep is left to future work. During Stage 3, buffer samples are replayed at one exemplar per four CST samples per mini-batch. The replayed distribution therefore spans the full T1 scale regime, intended to counteract the large-target gradient suppression examined in the S2→S3 analysis. To separate the effect of the herding selection from that of the stratification itself, SSH is compared both against naive fine-tuning and against a random-stratified control: the same buffer size and per-stratum quotas, but random selection within each stratum in place of herding (Section 4.6).

3.6 Datasets

Four datasets are used across the three stages. Anti-UAV-RGBT provides 318 paired RGB and IR video sequences with per-frame bounding box annotations on the IR stream. Across all splits it contains 293,209 UAV-present frames (frames in which a UAV is present, the `exist=1` flag); its training and validation splits (208,988 frames) have a scale distribution dominated by normal targets: 0.5% tiny, 28.0% small, 67.8% normal, and 3.7% large. Anti-UAV410 contains 428,703 UAV-present frames across all splits, with a train and validation scale distribution of 24.8% tiny, 25.9% small, 46.6% normal, and 2.7% large. CST Anti-UAV contains 208,221 UAV-present frames across all splits, with a near-inverted train and validation distribution: 87.9% tiny, 11.9% small, 0.3% normal, and 0% large. ARD100 provides 199,291 UAV-present ego-motion frames (202,411 total) used exclusively to pre-compute the motion difference masks for the YOLOMG temporal branch and is not part of the continual learning curriculum.

All datasets are accessed from their native storage formats during training to stay within the Snellius inode quota. Video-based datasets (Anti-UAV-RGBT, ARD100) are decoded on demand using cv2.VideoCapture; image-based datasets (Anti-UAV410, CST) are read from pre-extracted JPEG frames. Bounding box coordinates are normalised to $[0, 1]$ relative to a 640×512 resolution.

3.7 Evaluation Metrics

The Forgetting Measure (FM) is the primary metric for quantifying retention of earlier-task performance [2]:

$$\mathrm{FM} = \mathrm{mAP}_{T1}^{\text{after } T_k} - \mathrm{mAP}_{T1}^{\text{after } T1} \tag{2}$$

where both evaluations use the Anti-UAV-RGBT validation split. A negative FM indicates forgetting. The T1 ceiling is the Stage 1 best-checkpoint mAP (0.6725). Per-stratum mAP decomposes detection across the four size bins by bounding-box area (thresholds 256, 1024, and 4096 px^2, i.e. 16^2, 32^2, and 64^2). The scale-distribution figures, the per-stratum mAP, and the replay buffer all use this single area convention. Parameter drift between consecutive checkpoints is the mean cosine similarity across all 335 named tensors. Values near 1.0 indicate near-static weights, pointing to gradient starvation rather than bulk weight collapse.

For tracking evaluation on CST, Success Rate at Intersection over Union (IoU) 0.5 (SR@0.5), Precision Rate at 20 pixels (PR@20), and Identity Switch count (IDSW) are reported using the one-pass evaluation protocol [13].

3.8 Compute Infrastructure

All experiments are run on Snellius (SURF national supercomputer), using the A100 and H100 GPU partitions (per-stage allocation in Table 8). The full three-stage protocol—including the three Stage 2 seed replicates and the three Stage 3 variants—required on the order of one hundred hours of wall-clock training on 4-GPU nodes. This compute scale frames the scope of the study: no on-board or edge deployment is assumed, and whether resource-constrained hardware could support the continual (re)training studied here was not investigated; on-device continual learning is an open research problem in its own right [28, 29]. The Python environment uses Python 3.9, PyTorch 2.7.1, and CUDA 11.8. Random seeds are fixed at 42, 123, and 999 across all libraries to ensure reproducibility of multi-seed Stage 2 runs.

4 Results

4.1 Scale-Distribution Shift Across Domains

Before reporting per-stage results it is necessary to characterise how severely the three training domains differ in target size, since this difference is the primary driver of the forgetting observed in

Stage 3. Figure 2 shows the percentage of UAV-present frames in each of four UAV-specific size categories across all datasets. Size boundaries are by bounding-box area: tiny <256 px^2, small 256–1024 px^2, normal 1024–4096 px^2, large ≥4096 px^2 (i.e. 16^2 to 64^2). Standard COCO thresholds (32/96/192 px) do not suit UAV detection. A drone at 300 m typically subtends fewer than 32 pixels in thermal imagery. Under COCO conventions, then, most anti-UAV targets would count as “small” or “tiny” regardless of dataset.

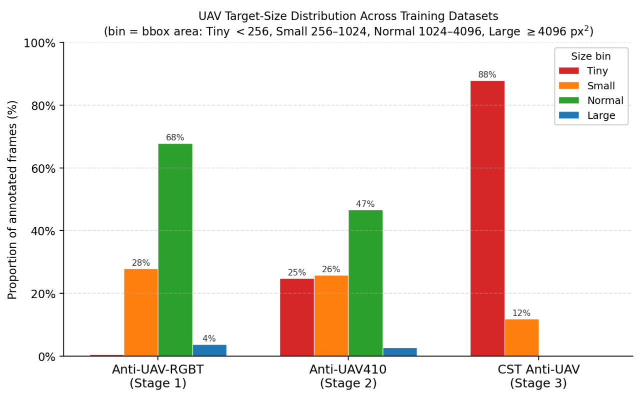

Figure 2: Scale distribution across the three training datasets (train + val splits, bbox area: tiny <256 px^2, small 256–1024 px^2, normal 1024–4096 px^2, large ≥4096 px^2). Anti-UAV-RGBT is dominated by normal targets (67.8%; 95.8% normal or small). CST Anti-UAV shows a near-complete inversion: 87.9% tiny; 99.8% tiny or small. All three datasets are thermal infrared, so the modality is held constant; the shift studied here is this scale/size distribution, not an arbitrary domain shift.

In Anti-UAV-RGBT (T1), 95.8% of UAV-present frames are normal or small (67.8% normal, 28.0% small) and only 3.7% are large; in CST Anti-UAV (T3), the normal and large strata are nearly absent (0.3% and 0%) and tiny targets dominate at 87.9%. Anti-UAV410 (T2) is intermediate (46.6% normal, 24.8% tiny), providing partial distribution overlap with T1. This inversion echoes Singh and Davis [37]: detectors trained on one scale regime perform poorly at a different scale. Their study was single-task; here the effect appears across sequential stages. The inversion motivates the gradient analysis that follows and the Scale-Stratified Herding design.

4.2 Catastrophic Forgetting Under Naive Sequential Training

Naive sequential training is expected to induce measurable catastrophic forgetting of T1 performance. The Forgetting Measure is:

$$\mathrm{FM}_k = \mathrm{mAP}_{T1}^{\text{after } T_k} - \mathrm{mAP}_{T1}^{\text{after } T1} \tag{3}$$

where mAP_{T1} is evaluated on the Anti-UAV-RGBT validation split and a value of zero indicates perfect retention.

Training the Stage 2 checkpoint on CST Anti-UAV for 19 epochs without any replay buffer drives T1 mAP@0.5 from 0.640 (the post-Stage-2 level) down to 0.068 at the best T3 checkpoint (epoch 3), and to 0.017 at the final epoch. Against the Stage 1 ceiling (0.6725) this is FM = −0.605. Of this, −0.572 is incurred in Stage 3 itself ($\mathrm{FM}_{\text{stage3}}$, relative to the Stage 2 baseline); the remaining −0.033 was already lost during Stage 2 distillation. The absolute post-Stage-3 forgetting (−0.605) is thus roughly 18 times the Stage 2 forgetting (FM = -0.033 ± 0.004 across three seeds), and the Stage 3 increment alone (−0.572) about 17 times. Scale-distribution shift therefore induces far more forgetting than cross-domain adaptation alone.

Table 1 shows the per-stratum breakdown. Large-target T1 mAP collapses to 0.000 by the end of training. The forgetting is concentrated precisely in the large-target stratum that CST Anti-UAV lacks entirely, while the tiny stratum is barely affected.

Table 1: Per-stratum T1 mAP@0.5 on Anti-UAV-RGBT across training stages. Stage 3 values are from the best naive checkpoint (epoch 3). After Sk is T1 mAP after Stage k; Δ is the net Stage 1 to Stage 3 change. Knowledge distillation retains every stratum through Stage 2—the large stratum is even slightly strengthened (0.461 to 0.494)—and the collapse occurs only at Stage 3. After S2 is the mean over three Stage 2 seeds (per-stratum std ≤ 0.006); After S1 (the Stage 1 ceiling) and After S3 (the naive run) are single checkpoints. Strata are defined by bounding-box area (thresholds 256/1024/4096 px^2).

Stratum	After S1	After S2	After S3	Δ
Tiny	0.009	0.018	0.001	−0.008
Small	0.579	0.559	0.060	−0.519
Normal	0.719	0.684	0.079	−0.640
Large	0.461	0.494	0.000	−0.461
Overall	0.673	0.640	0.068	−0.605

The per-epoch dynamics of the naive condition are shown in Figure 3 and detailed in Appendix D (Table 10). The large stratum falls to near-zero within the first training epoch and to 0.000 by epoch 1, never recovering. The normal stratum, the model’s primary detection strength after Stage 2 (mAP 0.681 on Anti-UAV410, Table 3), collapses from 0.251 at epoch 0 to 0.044 by epoch 7 and to 0.019 by the final epoch. T3 mAP peaks at epoch 3 (0.083) and then plateaus while T1 mAP continues declining; forgetting does not stop when the new task stops improving, a hallmark of catastrophic forgetting [24].

The forgetting is therefore measurable, large, and concentrated by stratum.

4.3 Knowledge Distillation During Stage 2 Training

Stage 2 training combines a standard detection loss with a knowledge distillation (KD) term to limit forgetting of Task 1 capability. The combined loss is:

$$L_{\text{total}} = L_{\text{det}} + \lambda_{\text{kd}} \cdot L_{\text{kd}}, \qquad \lambda_{\text{kd}} = 1.0 \tag{4}$$

where L_{det} is the standard detection loss on Anti-UAV410 labels and L_{kd} is the mean squared error between the student and the frozen Stage 1 teacher at prediction heads P3, P4, P5.

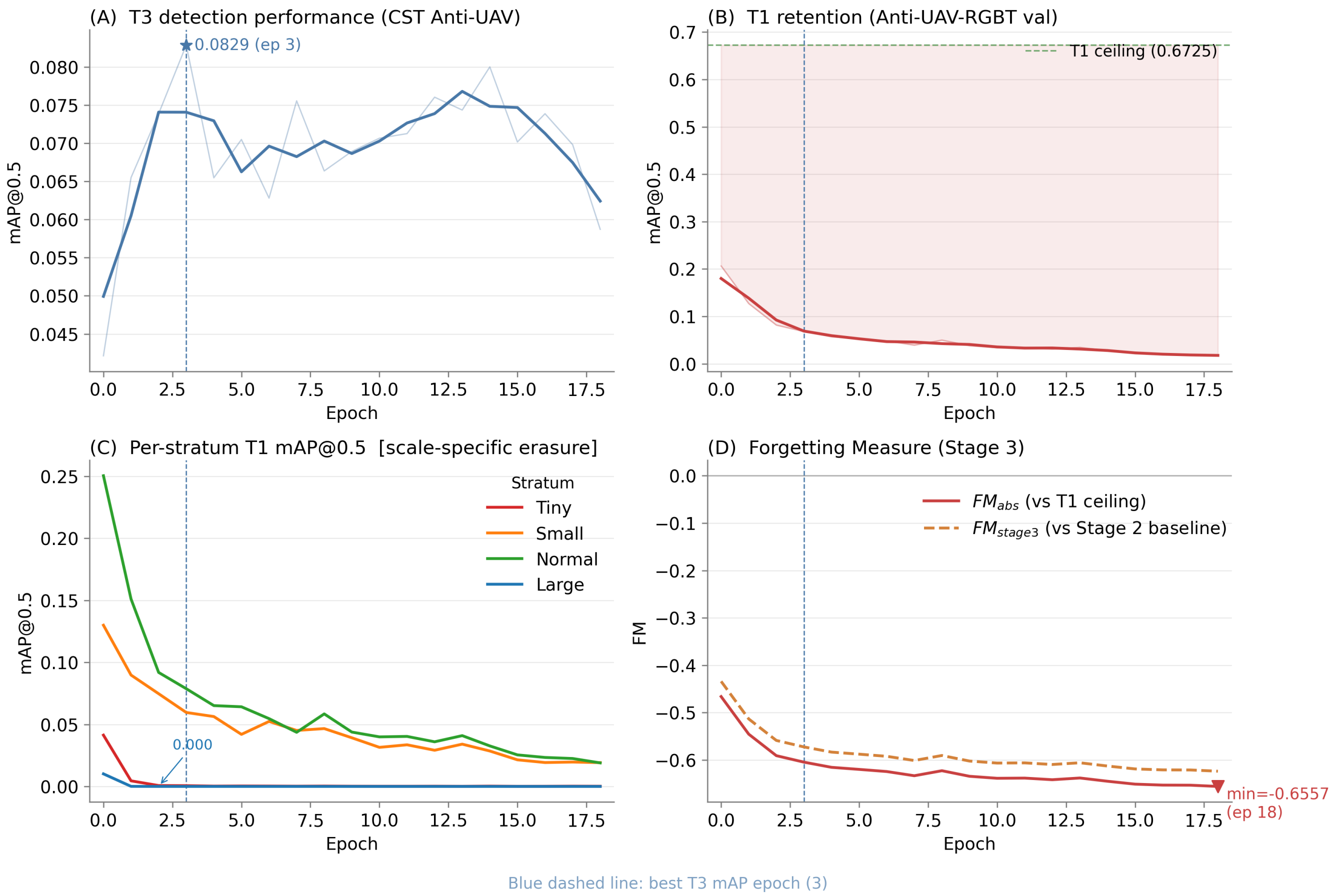

Figure 3: Stage 3 naive fine-tuning on CST Anti-UAV (19 epochs). (A) T3 detection mAP@0.5 on CST val; star marks the best checkpoint at epoch 3 (0.083). (B) T1 retention on Anti-UAV-RGBT val; shaded region shows the gap to the Stage 1 ceiling (0.6725). (C) Per-stratum T1 mAP@0.5: the large stratum (green) collapses to 0.000 at epoch 1, indicating scale-specific erasure. (D) Forgetting Measure: $\mathrm{FM_{abs}}$ (vs Stage 1 ceiling, dashed) and $\mathrm{FM_{stage3}}$ (vs Stage 2 baseline, solid); worst $\mathrm{FM_{abs}} = -0.656$ at epoch 18.

Table 2 reports Stage 2 aggregate results across three random seeds (42, 123, 999). KD limits forgetting to 2.8–3.7 percentage points of T1 mAP across the three seeds (FM -0.033 ± 0.004).

Table 2: Stage 2 aggregate results: KD fine-tuning on Anti-UAV410 ($\lambda_{\mathrm{kd}} = 1.0$, mean ± std across three seeds).

Metric	Value
T2 mAP@0.5 on Anti-UAV410 (best epoch)	0.428 ± 0.002
T1 mAP@0.5 retained (Anti-UAV-RGBT)	0.640
Forgetting Measure FM	−0.033 ± 0.004
T1 retention	95%
Best T2 epoch (seed 42)	16 / 32

A per-stratum breakdown indicates this retention is genuine, not an aggregate artefact (Table 1): every T1 stratum is preserved through Stage 2, and the large stratum, the one CST later erases, is even slightly strengthened (0.461 to 0.494). The 95% figure therefore reflects true preservation of the T1 representation, not mere overlap between Anti-UAV410 and T1.

Figure 4 shows the per-epoch Stage 2 trajectories across the three seeds. Epoch-level dynamics for seed 42 (Appendix D, Table 9) show two patterns. First, the kd/det ratio rises from 1.36 at epoch 0 to 2.62 at epoch 31: L_{det} declines faster than L_{kd} as the student adapts to Anti-UAV410, after which distillation provides a growing share of the gradient signal. Second, the FM deepens from −0.007 at epoch 0 to a trough of −0.046 at epoch 10, then partially recovers to −0.037 at the peak T2 epoch (16), illustrating the stability-plasticity trade-off directly: the checkpoint with the highest T2 mAP is not the one with the smallest forgetting.

The mean cosine similarity between Stage 1 and Stage 2 best checkpoints across all 335 named parameter tensors is 0.911 (per-seed values: 0.917, 0.921, 0.896), indicating high overall weight alignment. Restricting the measure to the 204 learnable (gradient-updated) tensors—excluding BatchNorm running-statistic buffers,

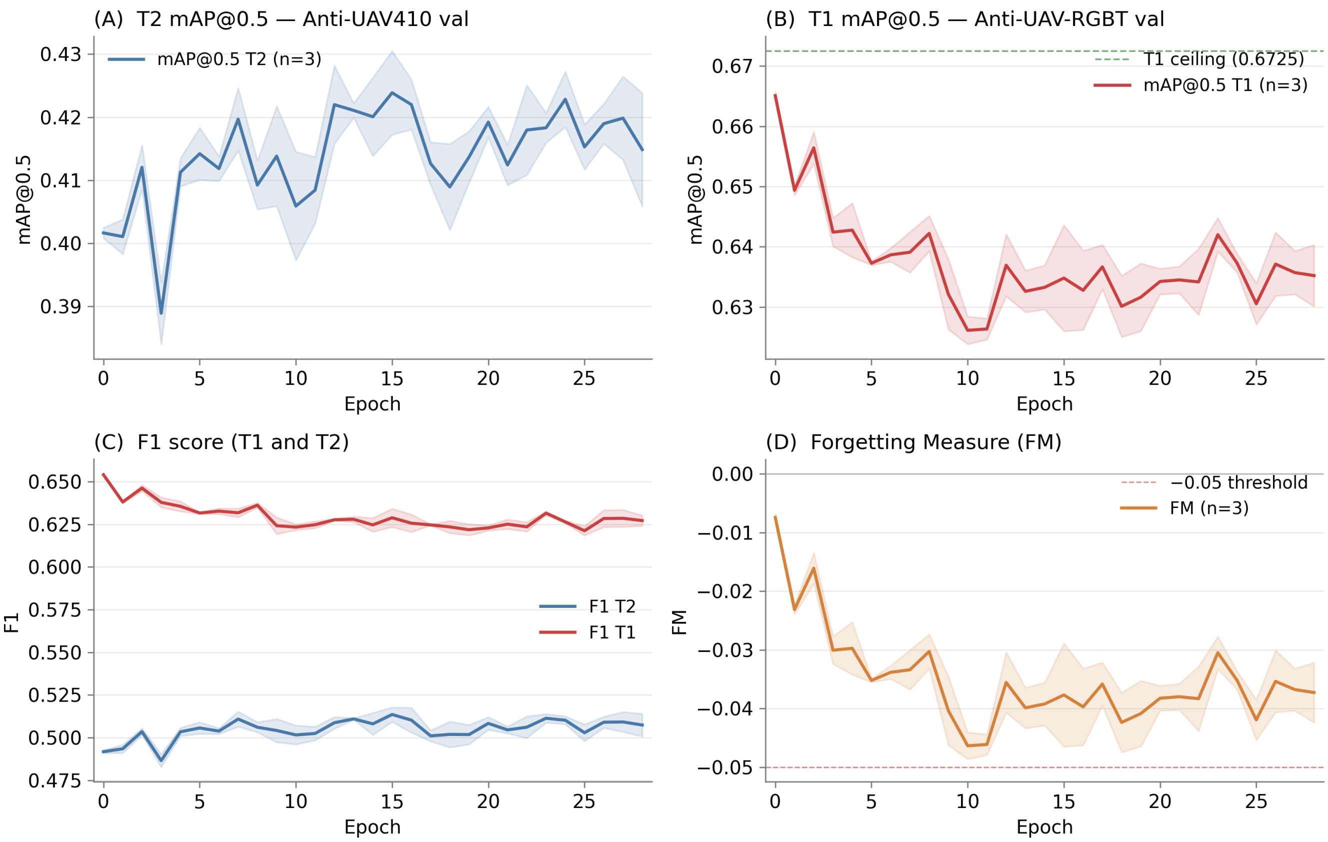

Figure 4: Stage 2 training dynamics across three seeds (n=3, mean with shaded ±1 std). (A) T2 mAP@0.5 on Anti-UAV410 val. (B) T1 mAP@0.5 on Anti-UAV-RGBT val; green dashed line is the Stage 1 ceiling (0.6725). (C) F1 score for T1 and T2 tasks. (D) Forgetting Measure per epoch; dashed horizontal marks the −0.05 boundary.

which adapt by forward-pass averaging rather than backpropagation—gives a comparable 0.935 (seed 42), so the alignment is a property of the trained weights rather than an artefact of the buffer count. The spatial attention module shows the largest relative drift, indicating the model adapted its low-level attention patterns for Anti-UAV410 while keeping backbone features close to the teacher.

Table 3 reports the scale-stratified mAP@0.5 on the Anti-UAV410 validation set (mean across three seeds). The Stage 2 model inherits the Stage 1 detector's strength at the normal stratum and retains moderate small-target performance, but is essentially unable to detect tiny targets (<256 px^2). This profile anticipates the scale-conditioned forgetting pattern of Stage 3: when fine-tuning on CST Anti-UAV (99.8% tiny/small, 0% large), the normal and large strata, almost absent from CST, receive little positive gradient signal from the new training distribution, a pattern probed directly in Section 4.5.

Together, these results show that KD at $\lambda_{\text{kd}} = 1.0$ reduces Stage 2 forgetting to FM $= -0.033 \pm 0.004$ (95% T1 retention), supported by both the rising kd/det ratio and the high inter-stage cosine similarity.

Table 3: Stage 2 scale-stratified mAP@0.5 on Anti-UAV410 validation set (mean across seeds 42, 123, 999). Normal-stratum performance (0.681) is the model's primary strength; tiny-stratum performance (0.001) is essentially zero throughout all stages.

Stratum	Area range	mAP@0.5 (mean)
Tiny	<256 px^2	0.001
Small	256–1024 px^2	0.278
Normal	1024–4096 px^2	**0.681**
Large	≥4096 px^2	0.276

4.4 Scale Shift and the Forgetting Pattern in Stage 3

CST Anti-UAV's training split contains 99.8% tiny/small targets and 0% large targets. When the Stage 2 checkpoint is fine-tuned on CST without replay, large-target T1 mAP falls to near-zero within the first training epoch and to 0.000 by epoch 1, while the tiny stratum barely changes (0.001 to 0.000). The collapse is not distributed uniformly: strata absent from T3 training data receive no gradient signal and collapse first.

To probe what underlies this pattern, inter-stage parameter drift is compared at both transitions (Table 4).

Table 4: Mean cosine similarity between successive stage checkpoints, computed over all 335 named tensors and, separately, over the 204 learnable (gradient-updated) tensors only, with BatchNorm running-statistic buffers excluded. Learnable-only values are for seed 42. Higher value means smaller weight change. FM is the Forgetting Measure relative to the Stage 1 ceiling.

Transition	Cosine (all)	Cosine (learn.)	FM
Stage 1 to Stage 2	0.911	0.935	−0.033
Stage 2 to Stage 3	0.967	0.987	−0.605

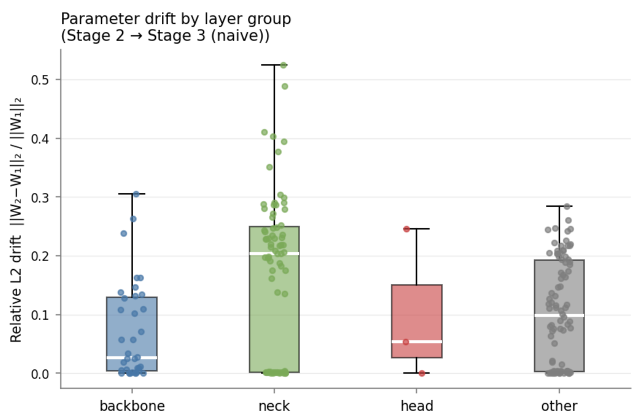

Figure 5: Relative L2 parameter drift (Stage 2 → Stage 3, naive) per layer group, restricted to the learnable (gradient-updated) tensors. The neck adapts most (median ≈0.20) and the backbone least (median ≈0.03); the three detection-head tensors show only modest drift (median ≈0.05). Excluding the BatchNorm running-statistic buffers removes the spurious head-layer drift seen when all parameters are pooled: the gradient-updated head weights barely move, consistent with starvation of the absent large-target stratum rather than active overwriting of the output layers.

The cosine similarity increases between transitions whether measured over all parameters (0.911 to 0.967) or over the learnable, gradient-updated tensors alone (0.935 to 0.987; Table 4). By either measure the trained weights moved *less* in Stage 3 than in Stage 2, relative to their starting point, yet FM is 18 times larger. This decoupling of weight-change magnitude from forgetting magnitude is consistent with gradient starvation [30]: large-target representations appear to receive insufficient gradient signal during Stage 3 training, as CST contains no large-target examples. The drift that does occur in Stage 3 sits not in the gradient-updated weights but in the BatchNorm running statistics (relative L2 ≈0.24 for `running_mean` and `running_var`, versus ≈0.12 for the learnable weights); among the learnable tensors the drift concentrates in the neck rather than the detection heads (Figure 5). The network tracks CST's input distribution through its normalisation buffers, while the trained weights stay close to their Stage 2 values. The unreinforced large-target output regime then degrades under weight decay and the small-target box-regression terms that dominate CST. The result is a stratum-specific collapse concentrated in the large stratum, the only stratum at 0.000 from epoch 1 onward, while the tiny stratum, already near zero in T1, is essentially unaffected.

Taken together, these results indicate that scale-distribution shift characterises the Stage 3 forgetting pattern at the stratum level. The evidence is consistent with a scale-conditioned gradient mechanism: inter-stage cosine similarity is higher at Stage 3 than at Stage 2 (0.987 vs 0.935 on the learnable tensors; 0.967 vs 0.911 on all parameters), yet forgetting is 18 times greater. Section 4.5 probes this mechanism directly.

4.5 Direct Gradient Probe at the Stage 2 → 3 Boundary

The cosine and BatchNorm evidence above is indirect. To examine the gradient signal directly, the Stage 2 checkpoint was probed *before* any Stage 3 optimisation: for 60 mini-batches each of CST (the Stage 3 distribution) and Anti-UAV-RGBT (T1), a forward and backward pass was run *without an optimiser step* (the network in training mode for loss computation, with BatchNorm layers frozen in evaluation mode so their running statistics were not perturbed), and the L_2 norm of the gradient reaching each detection head was recorded, together with the box-regression loss L_{box}. The probe was repeated for all three Stage 2 seeds; the values are stable across seeds and are reported as seed means in Table 5.

Table 5: Direct gradient probe at the Stage 2 → 3 boundary. Per-head gradient L_2 norm and box-regression loss L_{box} (mean over three Stage 2 seeds, 60 batches each, no optimiser step), under the CST distribution, the full T1 distribution, and T1 with only large (≥4096 px^2) ground truth retained. Heads are ordered by anchor size (P3: ∼3 px; P5: up to 26×14 px).

Quantity	CST (0% large)	RGBT (all)	RGBT (large only)
$\|\|g\|\|$ P3 (fine)	0.510	0.000	0.000
$\|\|g\|\|$ P4 (mid)	0.278	0.426	0.000
$\|\|g\|\|$ P5 (coarse)	0.388	0.615	0.227
L_{box}	0.096	0.020	0.001

Three observations follow. First, gradient flow is strongly scale-conditioned: a head receives gradient in proportion to the presence of size-matched targets. The clearest case is the finest head (∼3 px anchors), essentially silent on T1 ($||g|| \approx 0.000$, as RGBT contains almost no tiny (<256 px^2) targets) but strongly driven on CST ($||g|| \approx 0.51$, tiny-dominated). This is consistent with the starvation principle: a stratum absent from the training distribution receives no reinforcing signal.

Second, the detector's anchors are small by design (largest 26×14 px). A ≥64 px target is therefore weakly matched under the anchor ratio test, producing a near-zero positive box signal *even in*

T1 ($L_{\text{box}} \approx 0.001$ on large-only batches versus 0.020 on full T1). Large-target detection is therefore a fragile extrapolation rather than an anchor-matched capability, consistent with its rapid collapse once the distribution shifts. This capability was nonetheless real after Stage 1 (large-stratum mAP@0.5 = 0.461, Table 1): the probe explains why it is not maintained under CST, not that it was never present.

Third, CST drives a box-regression loss roughly five times larger than T1 ($L_{\text{box}} \approx 0.096$ vs 0.020): Stage 3 applies strong small-target localisation gradients to the shared heads while the large stratum receives no positive reinforcement, since CST's small targets keep even the coarsest head active ($\|g\| \approx 0.39$). The forgetting is thus better described as a *scale-conditioned gradient imbalance* (absence of positive large-target signal combined with dominant small-target gradients) than as starvation of a single dedicated head.

These measurements refine, rather than overturn, the cosine and BatchNorm evidence of Section 4.4: the learnable weights move little globally (Table 4), while the large-target output regime, never strongly anchored and now unreinforced, is not maintained. The probe supports the scale-conditioned account as a *candidate* mechanism; isolating its causal weight from the concurrent batch-size and learning-rate differences (Section 5) would require a controlled Stage 3 rerun and is left to future work.

4.6 Scale-Stratified Replay and Stage 3 Forgetting

Scale-Stratified Herding (SSH) addresses Stage 3 large-target forgetting by re-injecting large-target gradient signal through stratified exemplar replay, targeting the scale gap identified above.

The SSH buffer contains 300 exemplars partitioned equally across four size strata (75 per stratum). Within each stratum, greedy iCaRL herding [32] selects the 75 samples whose penultimate-layer embeddings are closest to the stratum mean, computed using the Stage 2 best checkpoint as the feature extractor:

$$x_k = \arg\min_{x \notin \mathcal{B}_{k-1}} \left\| \mu_s - \frac{1}{k}\left(\sum_{j<k} e(x_j) + e(x)\right) \right\|_2 \tag{5}$$

where $e(\cdot)$ is the global-average-pooled neck (P3) feature of the Stage 2 encoder and μ_s is the mean embedding of stratum s, matching the notation of Algorithm 1.

During Stage 3, each mini-batch mixes CST samples with replay exemplars from the buffer, restoring the large-target gradient signal CST cannot provide. This follows Deng et al. [5]: rehearsal is effective when the replayed distribution matches the original task in feature space. Stratifying by scale covers the full T1 scale range, whereas flat herding would replicate the training-split frequencies (Table 6), roughly two-thirds from the normal stratum alone, and leave the scarce strata uncovered.

The buffer is built using the Stage 2 seed-42 checkpoint as the feature extractor on the Anti-UAV-RGBT *training* split (148,368 UAV-present frames), kept disjoint from the validation set used to compute the FM so that replay introduces no train–evaluation overlap. Table 6 shows the per-stratum counts. The tiny stratum (1,091 eligible frames) is sampled at 6.9%, several times the large-stratum rate and 30–70× the small- and normal-stratum rates, deliberately over-representing tiny targets to protect that rare knowledge during Stage 3.

Table 6: SSH buffer composition (Anti-UAV-RGBT training split). Each stratum contributes exactly 75 exemplars regardless of size; the tiny stratum is over-represented (6.9% sampling rate) to protect rare tiny-target knowledge. Strata are by bounding-box area (thresholds 256/1024/4096 px^2).

Stratum	Eligible frames	Selected	Sampling rate
Tiny	1,091	75	6.9%
Small	43,560	75	0.2%
Normal	98,929	75	0.1%
Large	4,788	75	1.6%
Total	148,368	**300**	—

Trained from the Stage 2 seed-42 checkpoint, both replay conditions substantially mitigate the Stage 3 collapse (Table 7). At their best-T3 checkpoints (epoch 2), SSH reaches a Forgetting Measure of −0.311 and random-stratified replay −0.221, against the naive −0.605: each roughly halves the forgetting and holds large-target T1 mAP non-zero (0.079 and 0.129 respectively, versus 0.000), at a small cost to T3 detection (0.064 vs the naive 0.083 mAP), the expected stability–plasticity trade-off of replay.

Table 7: Stage 3 forgetting: naive fine-tuning versus random-stratified replay and Scale-Stratified Herding (SSH), each at its best-T3 checkpoint (single seed, seed 42). Both replay variants roughly halve the Forgetting Measure and keep the large stratum non-zero; herding does not improve on random selection.

Condition	T3 mAP	T1 mAP	FM	Large T1
Naive (ep. 3)	0.083	0.068	−0.605	0.000
Random-strat. (ep. 2)	0.064	0.451	−0.221	0.129
SSH (ep. 2)	0.064	0.362	−0.311	0.079

The random-stratified control isolates the source of the gain. Random selection within each stratum matches—indeed slightly exceeds—greedy herding, so the benefit appears to come from the *scale stratification* (balanced coverage of all size strata), not from the iCaRL-style herding selection. In this setting the herding component therefore seems unnecessary. Both runs are single-seed (seed 42) and short (Section 5); multi-seed, longer-schedule replication is left to future work.

4.7 Tracking Baseline on CST Anti-UAV

To establish a detection-as-tracking baseline, the Stage 3 naive checkpoint was evaluated on all 40 CST Anti-UAV validation sequences (39,055 ground-truth frames) under the one-pass protocol; the full precision and success plots are given in Appendix G. Tracking quality is low, as expected from the detector's collapse on CST: SR@0.5 = 0.012, AUC-SR = 0.039, PR@20 = 0.219, and 135 identity switches. The gap between PR@20 and SR@0.5 suggests the model often places its centre near the target but fails to size the box. This is the characteristic tiny-target failure, where a 1–2 px box error drops IoU below 0.5. These numbers set a baseline that a future uncertainty-aware gating mechanism would aim to improve upon.

4.8 Qualitative Detection Across Stages

Figure 6 shows detection output on a single Anti-UAV-RGBT validation frame under the three successive checkpoints. The Stage 1 and Stage 2 models both localise the target confidently, whereas the Stage 3 naive checkpoint produces no large-target detection, the qualitative counterpart of the stratum-specific erasure quantified above.

5 Discussion

5.1 Comparison with Prior Continual Learning Results

The Stage 2 Forgetting Measure of -0.033 ± 0.004 is consistent with what distillation-based continual learning achieves in general object detection. Shmelkov et al. [36] introduced an analogous distillation loss for incremental class detection on PASCAL VOC and reported near-zero forgetting on old classes when the distillation weight was tuned appropriately. Li and Hoiem [20] applied a similar output-distillation constraint for classification tasks and likewise found that a frozen teacher constrains forgetting to within a few percentage points under moderate domain shift. The 95% T1 retention observed here places KD in the same performance range. This offers early evidence that output-level distillation transfers to the thermal anti-UAV setting under a cross-dataset domain shift.

The Stage 3 result is harder to compare directly because most continual learning benchmarks study class-incremental learning on balanced datasets [40] rather than scale-distribution shift on a single class. The closest analogue is the general finding that forgetting is amplified when the new task distribution is maximally dissimilar from the old one [17]. The 18-fold increase from Stage 2 (-0.033) to Stage 3 (-0.605), with no change in the protection mechanism, fits this principle. The Anti-UAV410-to-Anti-UAV-RGBT gap is moderate (same modality, overlapping scales); the CST-to-Anti-UAV-RGBT gap is extreme (near-complete scale inversion, zero large-target overlap). What is novel is attributing this difference to a specific measurable quantity, the scale-distribution shift, rather than a qualitative description of task dissimilarity.

5.2 Gradient Starvation as a Candidate Forgetting Mechanism

The cosine similarity analysis in Table 4 produces a counterintuitive result: Stage 3 moves the weights less than Stage 2 does, relative to their starting point, yet Stage 3 forgetting is 18 times larger. This holds whether cosine is measured over all parameters (0.967 vs 0.911) or over the learnable, gradient-updated tensors alone (0.987 vs 0.935). The latter is cleaner: it excludes the BatchNorm running-statistic buffers, which are not updated by backpropagation. The standard account of catastrophic forgetting attributes performance loss to large weight changes that overwrite prior representations [17, 24]. Under that account, smaller weight changes should produce less forgetting, not more.

The gradient starvation framework [30] offers a plausible resolution to this contradiction. A representation that receives little or no gradient signal during new-task training is not actively overwritten by competing updates; the optimiser largely leaves it in place. This is consistent with the measurements: the gradient-updated weights move very little (learnable cosine 0.987), and the drift that does occur is concentrated not in those weights but in the BatchNorm running statistics (relative L2 ≈0.24 for `running_mean`/`running_var` versus ≈0.12 for the learnable weights). As the network re-estimates these normalisation statistics for CST's tiny-target input distribution, the activations of the now-unsupported large-target features can shift at test time even though the underlying weights barely change. The outcome is a collapse concentrated in exactly the strata absent from the new training distribution: as Table 1 shows, large-stratum mAP falls to near-zero within the first epoch and to 0.000 by epoch 1, while the tiny stratum is essentially unaffected.

A direct gradient probe (Section 4.5) both supports and qualifies this account. It indicates that gradient flow is scale-conditioned: a head receives signal only in proportion to the size-matched targets present, so a stratum absent from the data is left unreinforced (the finest head is silent on T1, which has almost no tiny ($<256\,\mathrm{px}^2$) targets, but strongly driven on tiny-dominated CST). It also shows, however, that the detector's small anchors (≤ 26 px) match ≥ 64 px targets only weakly, so the large-target pathway is fragile even under T1, and that CST drives a box-regression loss roughly five times larger than T1. The Stage 3 collapse is therefore most accurately described as a *scale-conditioned gradient imbalance* (the large stratum receives no positive reinforcement while strong small-target gradients dominate the shared heads): a refinement of, rather than a departure from, the starvation interpretation.

This has a direct implication for method choice. Regularisation methods such as EWC [17] penalise large weight changes, and gradient-projection methods such as OGD [7] and GPM [34] project new-task updates orthogonal to old-task directions; both target gradient *interference*: new-task updates overwriting old-task representations. The imbalance here is only partly of that kind: the shared heads do receive strong small-target gradients (Section 4.5), which a weight penalty could slow, but there is no large-target gradient to project or preserve, since that signal is absent from CST. Slowing the small-target updates cannot, on its own, restore a large-target signal the new distribution never supplies. This motivates a replay-based response that re-injects exemplars from the under-represented strata, supplying the missing gradient directly, the mechanism SSH provides.

5.3 Limitations

Several aspects of this study qualify the conclusions that can be drawn from it.

The optimisation setup is not perfectly matched: Stage 3 used batch 16 and learning rate 5×10^{-4} against batch 64 and 10^{-3} for Stage 2 (Table 8), so part of the larger Stage 3 forgetting could reflect these settings rather than scale shift alone. Two facts limit this concern: the forgetting is stratum-specific (a global change would not be stratum-specific), and large-target mAP collapses within the first epoch, before schedule differences accumulate. A Stage 3 rerun at the Stage 2 settings would isolate the effect.

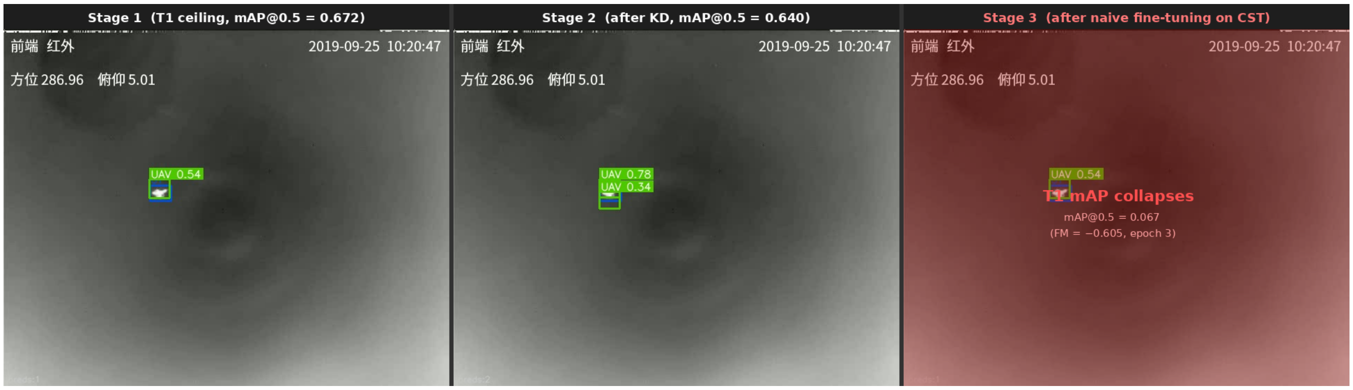

Figure 6: Detection output on the same Anti-UAV-RGBT validation frame (sequence 20190925_101846, clip 1_4, frame 0) under three successive checkpoints. Left: Stage 1 best checkpoint (T1 ceiling, mAP@0.5 = 0.6725). Centre: Stage 2 best checkpoint after knowledge distillation (mAP@0.5 = 0.640, FM = −0.033). Right: Stage 3 naive fine-tuning at epoch 3 (T1 mAP@0.5 = 0.068, FM = −0.605); the absence of large-target detections is in line with the gradient-starvation interpretation discussed above.

Generalisability is bounded by the data: all three datasets are predominantly DJI quadcopters in semi-controlled airspace, so the patterns may not transfer to fixed-wing UAVs, evasive targets, or very different backgrounds. The mechanism is also dataset-specific, depending on the particular Anti-UAV410-to-CST scale inversion; a smaller scale gap might shift the balance between weight drift and gradient imbalance.

On validity, the FM uses the Anti-UAV-RGBT validation split as its sole T1 reference, so a split biased toward easy sequences would understate forgetting. Stage 2 is replicated across three seeds (FM = −0.033 ± 0.004), but Stage 3 is single-seed (seed 42), which limits the precision of the Stage 3 estimate. We therefore claim a large, qualitatively robust effect rather than a seed-precise value: the Stage 3 FM (−0.605) is more than two orders of magnitude larger than the Stage 2 seed-to-seed spread (±0.004), the large-stratum collapse is immediate (0.000 by epoch 1) and permanent, and T1 mAP decays from 0.206 at epoch 0 to 0.017 at epoch 18 with no sustained recovery (transient upticks ≤0.011)—none of which is characteristic of a seed artefact, and all of which is consistent with continual-evaluation work showing that forgetting materialises within the first updates after a task switch [4]. Multi-seed replication of Stage 3 remains the first priority for future work.

Two design choices were not isolated. λ_{kd} was fixed at 1.0 by convention rather than tuned (equal weighting follows Li and Hoiem [20] and Shmelkov et al. [36]; tuning it would need a held-out FM signal, hence a full S1→S2→S3 run per value). And Stage 1 initialises from ImageNet weights, whose large-object prior means the post-Stage-1 large-stratum capability may be partly inherited.

The replay comparison is single-seed and short. Random-stratified replay slightly outperformed herding (FM −0.221 vs −0.311), suggesting the scale stratification rather than the herding selection drives the benefit; but with one seed and few epochs (five for SSH, three for the control) this ordering between the two replay variants is indicative rather than definitive. The buffer is drawn from the training split, disjoint from the validation set used for the FM, so replay introduces no train–evaluation overlap.

Finally, scalability was tested on a single architecture (YOLOMG); whether the finding and SSH transfer to other YOLO variants or transformer detectors is unknown, though the cosine diagnostic itself is architecture-agnostic. Relatedly, all conclusions concern the learning protocol rather than on-board deployment: training ran exclusively on HPC infrastructure (Section 3.8), and while inference-only use of YOLO-class detectors on embedded hardware is plausible, the feasibility of continual (re)training under edge constraints—an active research area in its own right [28, 29]—was neither assumed nor evaluated.

5.4 Value of the Research and Alternative Interpretations

This research offers three contributions to the continual learning literature. First, it sets out a concrete empirical protocol for measuring catastrophic forgetting in thermal anti-UAV detection across real operational datasets, producing the first quantified FM values for this domain. Second, it puts forward a scale-conditioned gradient imbalance (a form of gradient starvation) as a plausible mechanism under scale-distribution shift, diagnosed with a lightweight inter-stage cosine similarity that needs only the saved checkpoints. Third, it finds that scale-stratified replay (a buffer balanced across size strata) roughly halves the resulting forgetting. An ablation attributes this gain to the *stratification* (balanced coverage of size strata) rather than to the iCaRL herding selection: the two stratified variants perform comparably, whereas the no-replay baseline suffers the full collapse. This is consistent with recent imbalanced continual learning, where balancing the gradient contribution of under-represented groups, not the specific exemplar-selection rule, is the operative factor [11, 31]. The mechanism (well-represented scales degrading under a new scale-imbalanced distribution) is not specific to anti-UAV detection: the same imbalanced forgetting is reported in long-tailed continual recognition [26, 44], so both the diagnosis and the stratified-replay remedy should transfer to other scale- or long-tailed vision settings. Whether the mechanism is the dominant one, and whether the single-seed ordering between the replay variants holds, cannot be fully settled from the present evidence.

An alternative interpretation of the Stage 3 result is that the catastrophic collapse reflects not gradient starvation but a failure of the Stage 2 checkpoint to generalise well to Anti-UAV-RGBT in the first place: if the 95% T1 retention at Stage 2 is fragile rather than robust, even a small perturbation in Stage 3 could trigger a large FM drop. This interpretation is not supported by the data. The per-stratum breakdown (Table 1) shows every T1 stratum retained after Stage 2—the large stratum, which later collapses, is even strengthened (0.461 to 0.494)—and the per-epoch tracking (Table 9) shows stable T1 retention across all Stage 2 epochs. The collapse is Stage 3-specific and immediate, more consistent with the scale-conditioned gradient imbalance than with pre-existing fragility.

A second alternative is that the CST Anti-UAV annotations are noisier than Anti-UAV-RGBT annotations for sub-10-pixel targets, and that the detection loss instability from label noise rather than scale shift drives the forgetting. This cannot be ruled out from the available evidence, but the stratum-specific collapse pattern (large stratum affected most, tiny stratum barely affected) is not what noisy label training would produce: label noise would degrade all strata roughly uniformly, whereas gradient starvation predicts exactly the stratum-specific pattern observed.

A third reading questions whether Stage 3 measures forgetting at all, since the detector has almost no tiny-target capability before Stage 3. The two are separated by construction: forgetting is claimed only for strata whose capability was demonstrably acquired. The large stratum was learned in Stage 1 (mAP 0.461) and further strengthened in Stage 2 (0.494) before collapsing to 0.000 in Stage 3; this is forgetting of an acquired capability, matching the "similar-before, forget-after" signature documented for imbalanced continual learning [44]. The tiny stratum, by contrast, was never acquired (mAP ≈0 throughout), so no forgetting is claimed for it; its low pre-Stage-3 value reflects a plasticity limit of the small-anchor configuration rather than forgetting.

A note on YOLOMG's secondary input channel is warranted. Designed for motion-difference masks, it is held at a zero tensor throughout: Anti-UAV410 and CST lack the paired multi-frame sequences needed to pre-compute the masks, and they were not computed for Anti-UAV-RGBT within scope. Using Anti-UAV-RGBT's visible (RGB) stream as the Stage 1 secondary channel was deliberately avoided, since Stages 2 and 3 are purely thermal: the RGB channel would have to be zeroed mid-curriculum, adding a modality shift that confounds the scale shift. Holding it at zero throughout keeps the input modality constant, so the measured forgetting reflects scale shift alone.

The research serves a cautionary, diagnostic role. Naive sequential training across anti-UAV datasets appears unsuitable for deployment (FM = −0.605 is a 90% loss of T1 capability). Knowledge distillation works as an effective first-line defence for moderate domain shift. Scale-distribution shift, however, seems to call for a targeted replay-based response that existing regularisation methods are not designed to provide.

6 Conclusion

This research investigated catastrophic forgetting in a thermal anti-UAV detector (YOLOMG, operated as a single thermal-infrared stream) adapted sequentially across three heterogeneous benchmarks: Anti-UAV-RGBT, Anti-UAV410, and CST Anti-UAV.

Under sequential training, YOLOMG exhibits measurable catastrophic forgetting: naive fine-tuning on CST Anti-UAV produces a Forgetting Measure of −0.605 against the Stage 1 ceiling, of which −0.572 was incurred during Stage 3 alone, reducing Task 1 detection capability to roughly 10% of its post-Stage-1 level.

Knowledge distillation in Stage 2 limits forgetting to FM = -0.033 ± 0.004 across three seeds, corresponding to 95% T1 retention. Output-level distillation from a frozen teacher therefore appears to transfer to the thermal anti-UAV setting, acting as an effective first-line defence against forgetting under a thermal cross-dataset domain shift.

The Stage 3 results point to scale distribution shift as the amplifying factor. Moving from the moderate domain gap of Stage 2 to the extreme scale inversion of Stage 3 (CST Anti-UAV: 87.9% tiny, 0% large) increases forgetting by a factor of 18. Large-target mAP collapses to near-zero within the first Stage 3 epoch (0.000 by epoch 1) while the inter-stage cosine similarity over gradient-updated weights remains high at 0.987 (0.967 over all parameters), higher than the corresponding Stage 2 values (0.935 / 0.911). This dissociation argues against weight drift as the primary mechanism. It points instead to a scale-conditioned gradient imbalance, a refinement of gradient starvation [30], as a candidate. The large stratum receives no positive gradient signal from the new distribution, so its (weakly anchored) output regime erodes despite minimal change in the trained weights; the residual drift concentrates in BatchNorm running statistics that track the new input. A direct gradient probe (Section 4.5) is consistent with this scale-conditioned signal flow.

Scale-Stratified Herding (SSH) targets this failure mode: a 300-exemplar buffer selected by iCaRL-style greedy herding within four UAV size strata (75 exemplars per stratum) re-injects the missing gradient signal for underrepresented target scales. SSH roughly halves the Forgetting Measure (−0.605 → −0.311) and keeps large-target T1 mAP non-zero (0.079 vs 0.000), at a small cost to new-task detection. A random-stratified control performs at least as well (−0.221), indicating the gain comes from the scale stratification rather than the herding selection.

These conclusions are qualified by several limitations. Stage 3 was trained at a smaller batch size and lower learning rate than Stage 2, so part of the measured forgetting may reflect optimisation differences as well as scale shift. The gradient-starvation finding also depends on the particular scale inversion between Anti-UAV410 and CST Anti-UAV and may not generalise to dataset pairs with smaller scale gaps. Finally, the Stage 3 and replay results rest on a single seed and short (few-epoch) runs, so the small margin between the two replay variants is indicative rather than definitive.

The primary contribution of this research is empirical: it provides the first quantified forgetting protocol for thermal anti-UAV detection, and suggests that naive sequential training across operational datasets is unsuitable for deployed C-UAV systems (FM = −0.605, 90% capability loss). Three directions for future work follow from these findings. Most immediately, the single-seed, few-epoch replay results should be replicated across multiple seeds and longer schedules to confirm the effect size and the currently narrow margin between herding and random replay. Comparing the stratified-replay response against regularisation methods such as EWC [17] and GEM [22] would clarify whether replay or regularisation better counters the imbalance under scale shift. Finally, integrating the detector with uncertainty-aware tracking [35, 49], improving on the tracking baseline of Section 4.7, would extend the framework to the full detection-to-tracking pipeline, addressing the complete C-UAV operational scenario.

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Appendix A Declaration on the Use of Generative AI

During the preparation of this work, the author used Claude (Anthropic) in a supporting capacity. It was used to format and restructure LaTeX source, debug code and analysis scripts, check grammar and spelling, and improve language flow and phrasing. All experimental design, data collection, result interpretation, and scientific conclusions are the author's own. The author remained in the loop throughout (no AI output was used without critical review and revision) and takes full responsibility for the content of this research.

Appendix B Dataset Statistics

Figure 7 reports the per-split UAV-present frame counts for each dataset, and Figure 8 the corresponding target-visibility rates.

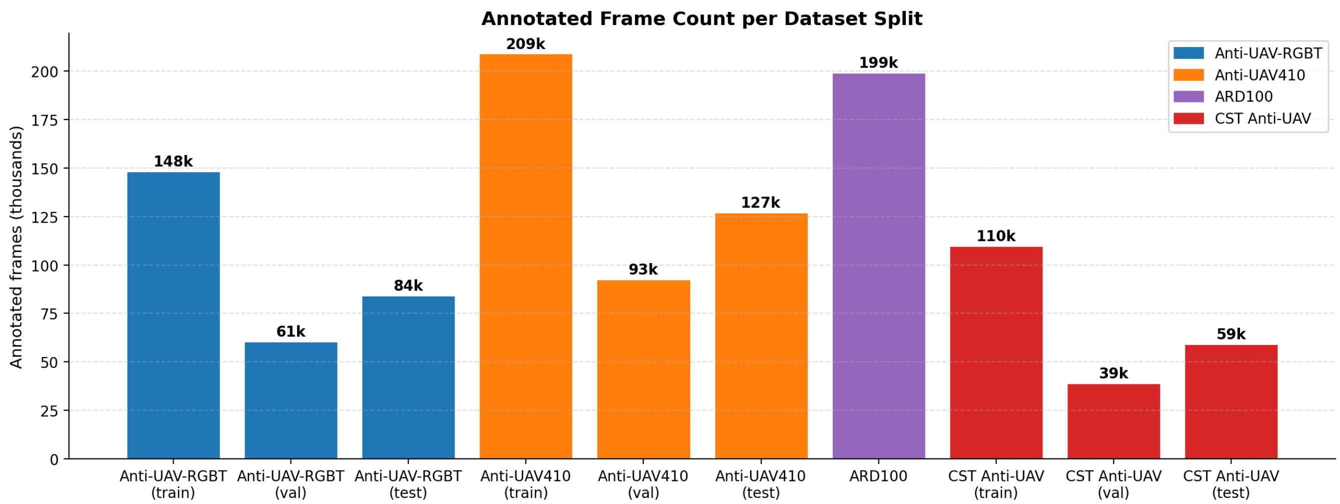

Figure 7: Per-split counts of UAV-present frames—frames in which a UAV is present (the `exist=1` flag), labelled 'annotated' on the figure axis. Anti-UAV410 is the largest (428,703 across train/val/test), followed by Anti-UAV-RGBT (293,209). CST Anti-UAV (208,221) and ARD100 (199,291) are comparable in size; ARD100 is used only for motion-mask pre-computation and is not part of the continual learning curriculum.

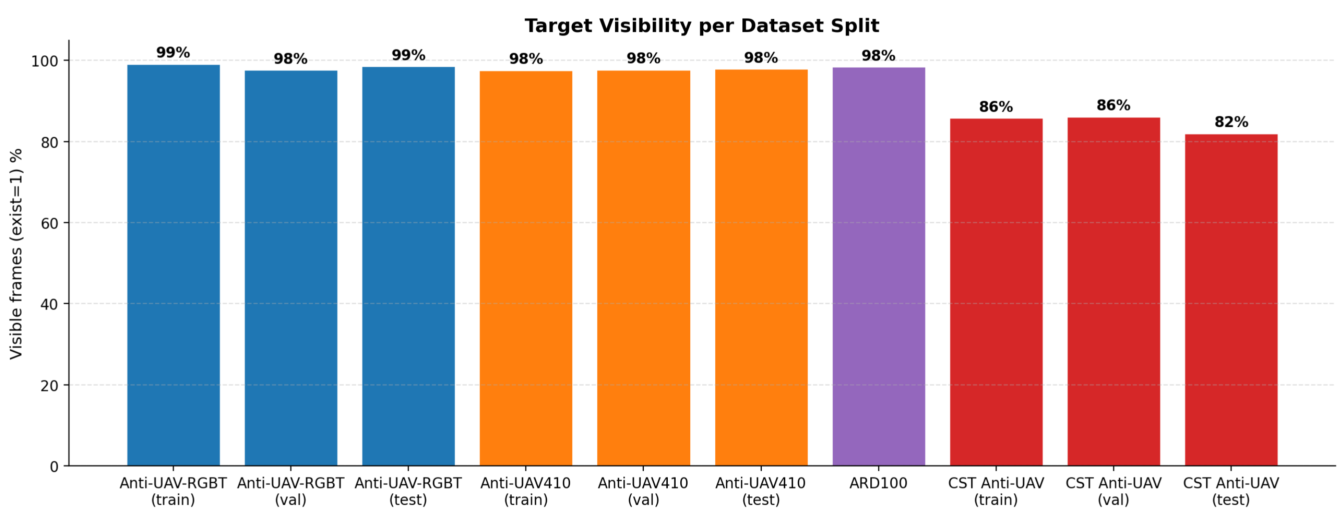

Figure 8: Target visibility per dataset split. A frame is marked visible if it contains an actively flying UAV (the `exist=1` flag); occluded or out-of-frame instances are marked absent. Anti-UAV-RGBT has the highest visibility rate, CST Anti-UAV the lowest, consistent with the difficulty of detecting sub-16-pixel targets in cluttered thermal scenes.

Appendix C Experimental Configuration

Table 8 summarises the hyperparameters used across all three training stages. All stages use Nesterov SGD with cosine learning rate decay and the same YOLOMG architecture (318 layers, 335 named parameter tensors, single UAV class).

Table 8: Training configuration per stage. Stage 2 is run with three seeds; Stages 1 and 3 use seed 42 only. Epochs are 0-indexed (epoch 0 is the first).

	Stage 1	Stage 2	Stage 3
Dataset	Anti-UAV-RGBT (IR)	Anti-UAV410	CST Anti-UAV
Objective	Supervised det.	KD + det.	Naive / replay det.[†]
Epochs	100 (early stop)	50 (early stop)	naive: ran 19, best ep. 3; SSH: ran 5, best ep. 2; random: ran 3, best ep. 2
Best checkpoint	epoch 49	ep. 16/15/27 (seeds 42/123/999)	see Epochs row
Optimiser	SGD (Nesterov)	SGD (Nesterov)	SGD (Nesterov)
Initial LR	10^{-2}	10^{-3}	5×10^{-4}
LR schedule	Cosine decay	Cosine decay	Cosine decay
Weight decay	5×10^{-4}	5×10^{-4}	5×10^{-4}
Momentum	0.937	0.937	0.937
Batch size	64 (4×A100)	64 (4×H100)	16 (4×A100)
λ_{kd}	—	1.0	—
Seeds	42	42, 123, 999	42
Input ch. 1	IR frame	IR frame	IR frame
Input ch. 2	zeros	zeros	zeros

[†] Three Stage 3 variants were run: naive (no replay), SSH (scale-stratified herding replay), and random-stratified replay. All share the same hyperparameters; replay variants additionally use replay weight = 4.0, replay ratio = 25%, buffer size = 300 (75 per stratum).

Appendix D Additional Training Dynamics

Table 9 reports the epoch-level Stage 2 dynamics (seed 42) and Table 10 the per-epoch Stage 3 naive dynamics; both are discussed in the main text (Sections 4.3 and 4.2).

Table 9: Epoch-level Stage 2 results (seed 42, 32 epochs). kd/det is the ratio of L_{kd} to L_{det}, averaged over batches. Epoch 16 is the peak T2 checkpoint. FM values here are seed 42's Stage 2 training-time evaluation (e.g. -0.037 at epoch 16). The post-Stage-2 baseline used for the headline Forgetting Measure and as the Stage 3 starting point is a clean re-evaluation of this checkpoint (T1 mAP 0.640, FM -0.033); the $\sim$0.004 gap reflects training-time versus re-evaluation differences. The three-seed mean Stage 2 FM is -0.033 ± 0.004.

Epoch	L_{det}	L_{kd}	kd/det	T2 mAP	FM
0	0.0580	0.0791	1.36	0.403	−0.007
5	0.0355	0.0768	2.16	0.419	−0.035
10	0.0317	0.0739	2.33	0.406	−0.046
15	0.0293	0.0712	2.43	0.417	−0.042
16	0.0289	0.0705	2.44	0.426	−0.037
20	0.0273	0.0677	2.48	0.419	−0.041
25	0.0256	0.0648	2.54	0.415	−0.044
31	0.0233	0.0612	2.62	0.422	−0.037

Appendix E Preprocessing Pipelines

Motion Mask Generation (mask32)

YOLOMG's second input channel is designed to carry a pixel-level motion difference map (mask32), computed from five consecutive frames after global motion compensation (GMC). GMC uses the Enhanced Correlation Coefficient (ECC) algorithm [6] to estimate the camera's ego-motion between frames, warp earlier frames to align with the current frame, and then compute the frame-difference residual. The residual retains only object-relative motion, suppressing camera shake.

The five-frame sliding window operates as follows. Let frames be indexed $t-2, t-1, t, t+1, t+2$. The FD5 scheme uses three non-adjacent frames ($t-2$, t, and $t+2$) to compute the motion difference after ECC-based alignment. For Stage 1 (Anti-UAV-RGBT) and all purely thermal datasets (Anti-UAV410, CST Anti-UAV), motion masks were not available within the scope of this study, so the second input channel is set to

Table 10: Stage 3 naive condition — per-epoch results. Pre-training T1 mAP (before any Stage 3 gradient steps) = 0.640. T1 ceiling = 0.6725. Bold marks the best T3 checkpoint (epoch 3). Large-stratum T1 mAP reaches 0.000 at epoch 1 and never recovers; the small and normal strata fall steeply in parallel (from 0.559 and 0.684 post-Stage-2, Table 1, to 0.060 and 0.079 at the best checkpoint).

Ep.	T3 mAP	T1 mAP	FM_{abs}	FM_{stage3}	Tiny	Small	Normal	Large
pre	—	0.640	−0.033	0.000	—	—	—	—
0	0.042	0.206	−0.466	−0.434	0.041	0.130	0.251	0.010
1	0.066	0.127	−0.545	−0.513	0.005	0.090	0.151	0.000
2	0.074	0.082	−0.591	−0.559	0.001	0.075	0.092	0.000
3	**0.083**	**0.068**	**−0.605**	**−0.572**	**0.001**	**0.060**	**0.079**	**0.000**
4	0.066	0.057	−0.615	−0.583	0.000	0.056	0.065	0.000
5	0.071	0.053	−0.620	−0.588	0.000	0.042	0.064	0.000
6	0.063	0.048	−0.624	−0.592	0.000	0.052	0.055	0.000
7	0.076	0.039	−0.633	−0.601	0.000	0.045	0.044	0.000
18	0.062	0.017	−0.656	−0.623	0.000	0.019	0.019	0.000

a zero tensor throughout. This means the motion branch of YOLOMG contributes no signal during training and the model functions as a single-stream thermal appearance detector.

Dataset Loading Strategy

To avoid inode quota exhaustion on Snellius (`/projects/prjs2041/`), no intermediate files are written to disk. Each dataset is read in its native format inside a custom PyTorch `Dataset` class. Table 11 summarises the loading strategy per dataset.

Table 11: Native-format loading strategy per dataset. No intermediate converted files are written; all format-specific logic is encapsulated inside the `Dataset` class.

Dataset	Frame source	Annotation source	Extra files
Anti-UAV-RGBT	`cv2.VideoCapture` (`.mp4`)	`infrared.json`	None
Anti-UAV410	`cv2.imread` (pre-extracted JPG)	`IR_label.json`	None
CST Anti-UAV	`cv2.imread` (pre-extracted JPG)	`gt.txt` + `IR_label.json`	None
ARD100	`cv2.VideoCapture` (`.mp4`)	Pascal VOC XML per frame	None

Frame Extraction

Anti-UAV-RGBT stores sequences as `.mp4` video files with per-frame JSON annotations. Frames are decoded on demand using `cv2.VideoCapture` with frame-index seeking (or pre-extracted as JPEG files to eliminate random-seek overhead on NFS mounts). Only the infrared stream (`infrared.mp4`) is loaded; the paired RGB stream (`visible.mp4`) is ignored entirely. Anti-UAV410 and CST Anti-UAV store pre-extracted JPEG frames, which are read directly.

Scale Stratification

Bounding box scale stratification is applied consistently across all scripts using bounding-box area in pixels. A normalised box (w_n, h_n) is first converted to pixel dimensions at the canonical resolution 640×512:

$$a = (w_n \times 640) \times (h_n \times 512)$$

The four strata are:

- tiny: $a < 256\,\mathrm{px}^2$ $(= 16^2)$
- small: $256 \le a < 1024\,\mathrm{px}^2$ $(16^2–32^2)$
- normal: $1024 \le a < 4096\,\mathrm{px}^2$ $(32^2–64^2)$
- large: $a \ge 4096\,\mathrm{px}^2$ $(\ge 64^2)$

These thresholds are used in the evaluation script (`eval_full_analysis.py`), the scale distribution plot (`plot_scale_distribution.py`), and the herding buffer builder (`build_herding_buffer.py`).

Appendix F Pseudocode

Algorithm 1: Scale-Stratified Herding (SSH)

Algorithm 1 formalises the SSH buffer construction. The key departure from standard iCaRL herding is step 2, which partitions the T1 data by size stratum before applying greedy herding, ensuring equal representation of all four scale categories regardless of their frequency in the source dataset.

Algorithm 1 Scale-Stratified Herding (SSH)

Require: T1 reference set $\mathcal{D}_1$ (Anti-UAV-RGBT training split, 148,368 UAV-present frames), Stage 2 encoder f_{θ_2}, exemplars per stratum $K = 75$, strata $\mathcal{S} = \{\text{tiny, small, normal, large}\}$

Ensure: Exemplar buffer $\mathcal{M}$ with $|\mathcal{M}| = 4K = 300$

1: **Extract embeddings.** For each $x \in \mathcal{D}_1$:

2: $e(x) \leftarrow \text{GlobalAvgPool}\left(f_{\theta_2}^{P3}(x)\right)$ ▷ neck P3 feature, 128-dim

3: Assign stratum $s(x)$ by bounding-box area (thresholds: 256, 1024, 4096 px^2)

4: $\mathcal{M} \leftarrow \emptyset$ **for** *each stratum* $s \in \mathcal{S}$ **do**

5:

end

$\mathcal{D}_s \leftarrow \{x \in \mathcal{D}_1 : s(x) = s\}$

6: $\mu_s \leftarrow \frac{1}{|\mathcal{D}_s|} \sum_{x \in \mathcal{D}_s} e(x)$ ▷ full stratum mean

7: selected $\leftarrow [\,]$; $\bar{e} \leftarrow \mathbf{0}$ **for** $k = 1, \ldots, K$ **do**

8:

end

$j^* \leftarrow \arg\min_{j \notin \text{selected}} \left\| \frac{\bar{e} \cdot (k-1) + e(x_j)}{k} - \mu_s \right\|_2$

9: selected.append(j^*)

10: $\bar{e} \leftarrow \frac{\bar{e} \cdot (k-1) + e(x_{j^*})}{k}$ ▷ update running mean

11:

12: $\mathcal{M} \leftarrow \mathcal{M} \cup \{x_j : j \in \text{selected}\}$

13:

14: **return** $\mathcal{M}$

Algorithm 2: Stage 2 Knowledge Distillation

Algorithm 2 shows the Stage 2 training loop. The teacher is initialised from the Stage 1 checkpoint and fully frozen before training begins. Both the student and teacher receive the same IR frame; the student is additionally supervised by Anti-UAV410 ground-truth annotations.

Algorithm 2 Stage 2: Knowledge Distillation Training

Require: Stage 1 checkpoint θ_T, Anti-UAV410 training set $\mathcal{D}_2$, distillation weight $\lambda_{\text{kd}} = 1.0$, epochs E

Ensure: Stage 2 student checkpoint θ_S^*

$\theta_S \leftarrow \theta_T$ ▷ initialise student from teacher weights

Freeze all parameters of θ_T; set f_{θ_T} to eval mode

for *epoch* $e = 1, \ldots, E$ **do**

end

each mini-batch $(x, y) \in \mathcal{D}_2$ ▷ x: IR frame, y: GT boxes

$\mathbf{p}_S \leftarrow f_{\theta_S}(x, \mathbf{0})$ ▷ student raw grids at P3, P4, P5

$\mathbf{p}_T \leftarrow f_{\theta_T}(x, \mathbf{0})$ ▷ teacher grids, no gradient

$\mathcal{L}_{\text{det}} \leftarrow \text{YOLOLoss}(\mathbf{p}_S, y)$

$\mathcal{L}_{\text{kd}} \leftarrow \frac{1}{3} \sum_{p \in \{P3, P4, P5\}} \text{MSE}(\mathbf{p}_S^{(p)}, \mathbf{p}_T^{(p)})$

$\mathcal{L} \leftarrow \mathcal{L}_{\text{det}} + \lambda_{\text{kd}} \cdot \mathcal{L}_{\text{kd}}$

Update θ_S via SGD on $\nabla_{\theta_S} \mathcal{L}$

Evaluate student on Anti-UAV410 val and Anti-UAV-RGBT val

$\theta_S^* \leftarrow \arg\max_e \ \text{mAP@0.5}_{T2}(\theta_S^{(e)})$

return θ_S^*

Algorithm 3: Forgetting Measure Computation

Algorithm 3 defines the Forgetting Measure used throughout this research. The T1 ceiling is fixed once after Stage 1 and reused as the denominator baseline for all subsequent stages.

Algorithm 3 Forgetting Measure (FM)

Require: Stage 1 checkpoint θ_1, Stage k checkpoint θ_k, Anti-UAV-RGBT validation split $\mathcal{D}_1^{\text{val}}$
Ensure: Forgetting Measure FM

$m_1 \leftarrow \text{mAP@0.5}\big(f_{\theta_1}, \mathcal{D}_1^{\text{val}}\big)$ ▷ T1 ceiling, fixed at 0.6725
$m_k \leftarrow \text{mAP@0.5}\big(f_{\theta_k}, \mathcal{D}_1^{\text{val}}\big)$ ▷ T1 retention after Stage k
$\text{FM} \leftarrow m_k - m_1$ ▷ negative = forgetting
return FM

Appendix G Stage 3 Tracking Evaluation

Figures 9–11 report the one-pass evaluation (OPE) tracking results for the Stage 3 best checkpoint on the CST Anti-UAV validation set, using the protocol of Huang et al. [13]. The precision plot (Figure 9) shows the proportion of frames whose predicted centre falls within a threshold pixel distance of the ground-truth centre; the success plot (Figure 10) shows the proportion of frames with IoU overlap above threshold. Figure 11 breaks success rate at IoU 0.5 (SR@0.5) down by sequence, revealing the substantial per-sequence variance that is not visible in the aggregate mAP.

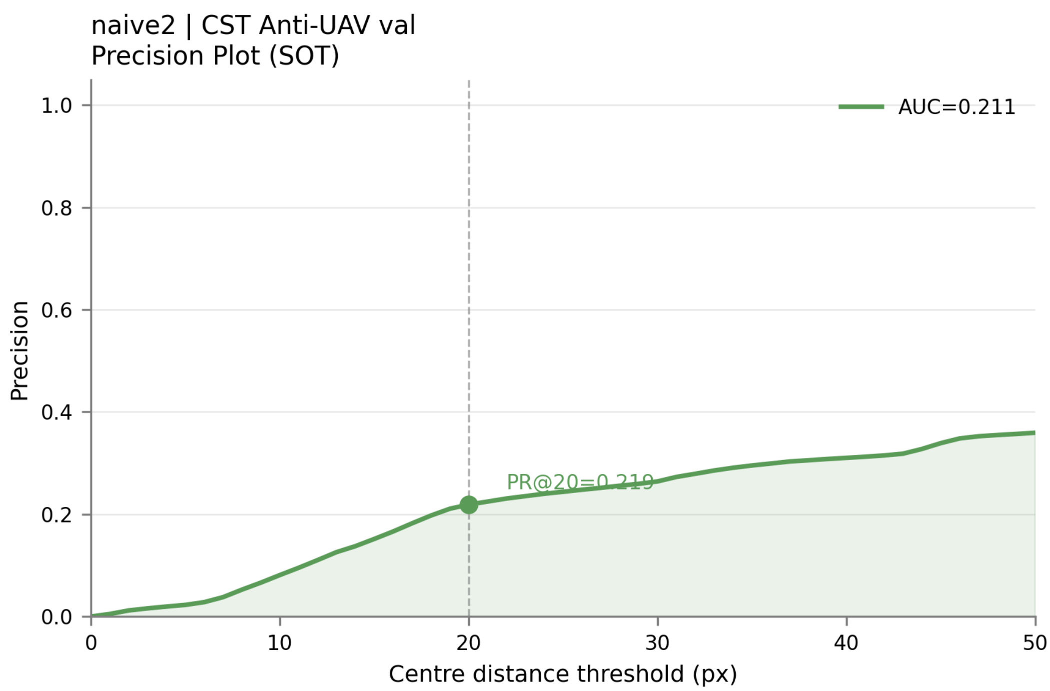

Figure 9: Precision plot on CST Anti-UAV validation set (Stage 3 best checkpoint, epoch 3). The representative score at threshold 20 px (PR@20) is reported in the main tracking evaluation table.

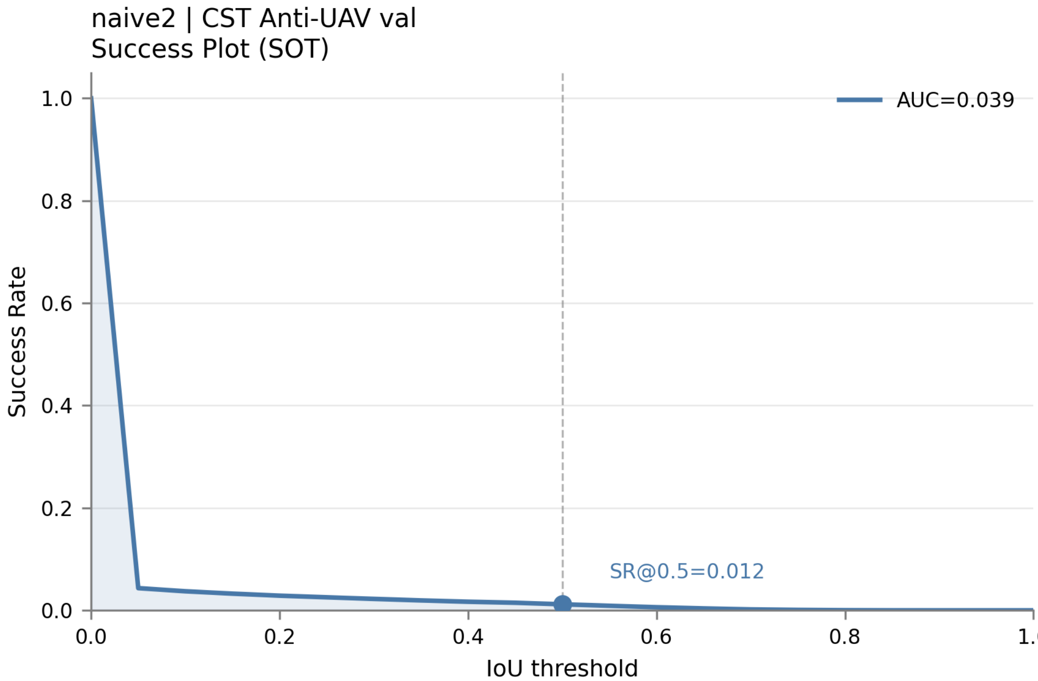

Figure 10: Success plot on CST Anti-UAV validation set (Stage 3 best checkpoint, epoch 3). The area under this curve (AUC) provides an IoU-threshold-independent summary of tracking quality.

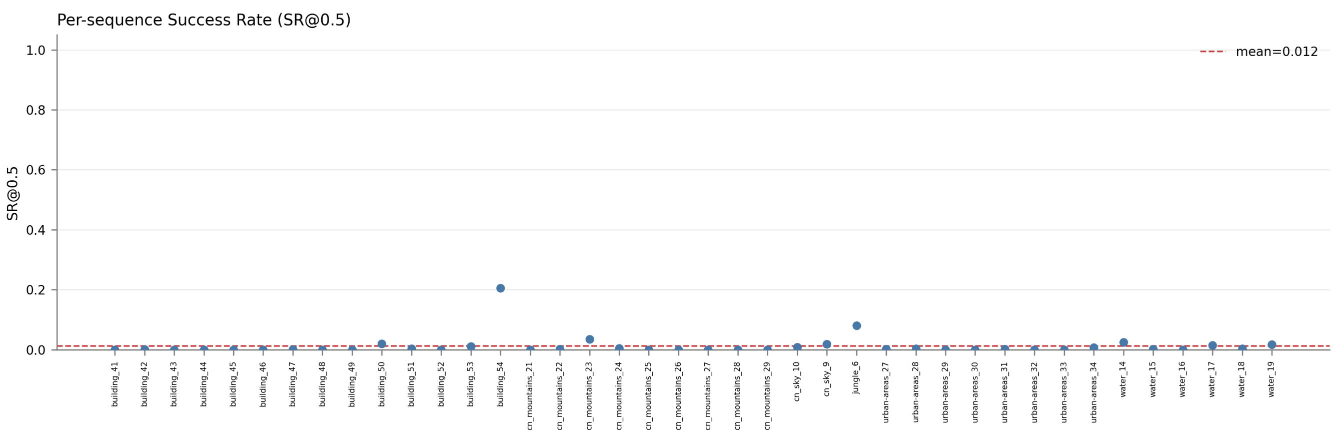

Figure 11: Per-sequence success rate at IoU 0.5 (SR@0.5) on CST Anti-UAV validation set. Sequences with predominantly tiny targets (<256 px^2) show the lowest SR@0.5, consistent with the near-zero tiny-stratum mAP reported in Table 1.

Appendix H Compute Environment

All training and analysis jobs were executed on the Snellius HPC cluster (SURF, the Netherlands). The software environment was a dedicated Conda environment (`uav_master`, Python 3.9) created under `/projects/prjs2041/` using a hybrid strategy: Conda created the Python sandbox; all packages were installed with `pip` to avoid the Conda dependency solver freezing on login nodes.

Library versions.

Package	**Version**
Python	3.9
torch	2.7.1+cu118
torchvision	0.22.1+cu118
torchaudio	2.7.1+cu118
numpy	2.0.2
opencv-python	(latest at setup)
scipy	(latest at setup)
pandas	(latest at setup)
PyYAML	(latest at setup)
tqdm	(latest at setup)

Standard session startup. Every new session on Snellius requires the following module sequence before running any script:

```
module load 2023
module load Miniconda3/23.5.2-0
source activate uav_master
# Always use the absolute interpreter path:
/home/knguyen1/.conda/envs/uav_master/bin/python script.py
```

This is necessary because Snellius uses a hierarchical Lmod system where Conda is hidden until the correct software stack year is loaded, and because VS Code's Open OnDemand interface may intercept the `python` alias without this explicit activation.

Appendix I Training Run Log

Table 12 records the training and analysis jobs that produced the results reported in this work, provided for reproducibility.

Table 12: Snellius jobs underlying the reported results.

Job ID	Stage	Notes
23297129	S1	Supervised training; best mAP@0.5 = 0.6725 at ep. 49
23349180	S2	KD fine-tuning, seed 42; FM = −0.037
23349181	S2	KD fine-tuning, seed 123; FM = −0.028
23349182	S2	KD fine-tuning, seed 999
23394054	S3	Naive baseline; FM = −0.605, best at ep. 3
23381446	S2	Multi-seed confidence-interval aggregation
23391452	S3	Replay buffer construction (herding + random)
24194142	S3	SSH replay training, seed 42; FM = −0.311, best at ep. 2
24197035	S3	Random-stratified replay training, seed 42; FM = −0.221, best at ep. 2
24144797	—	Gradient probe at the Stage 2 $\longrightarrow$ 3 boundary (Section 4.5)